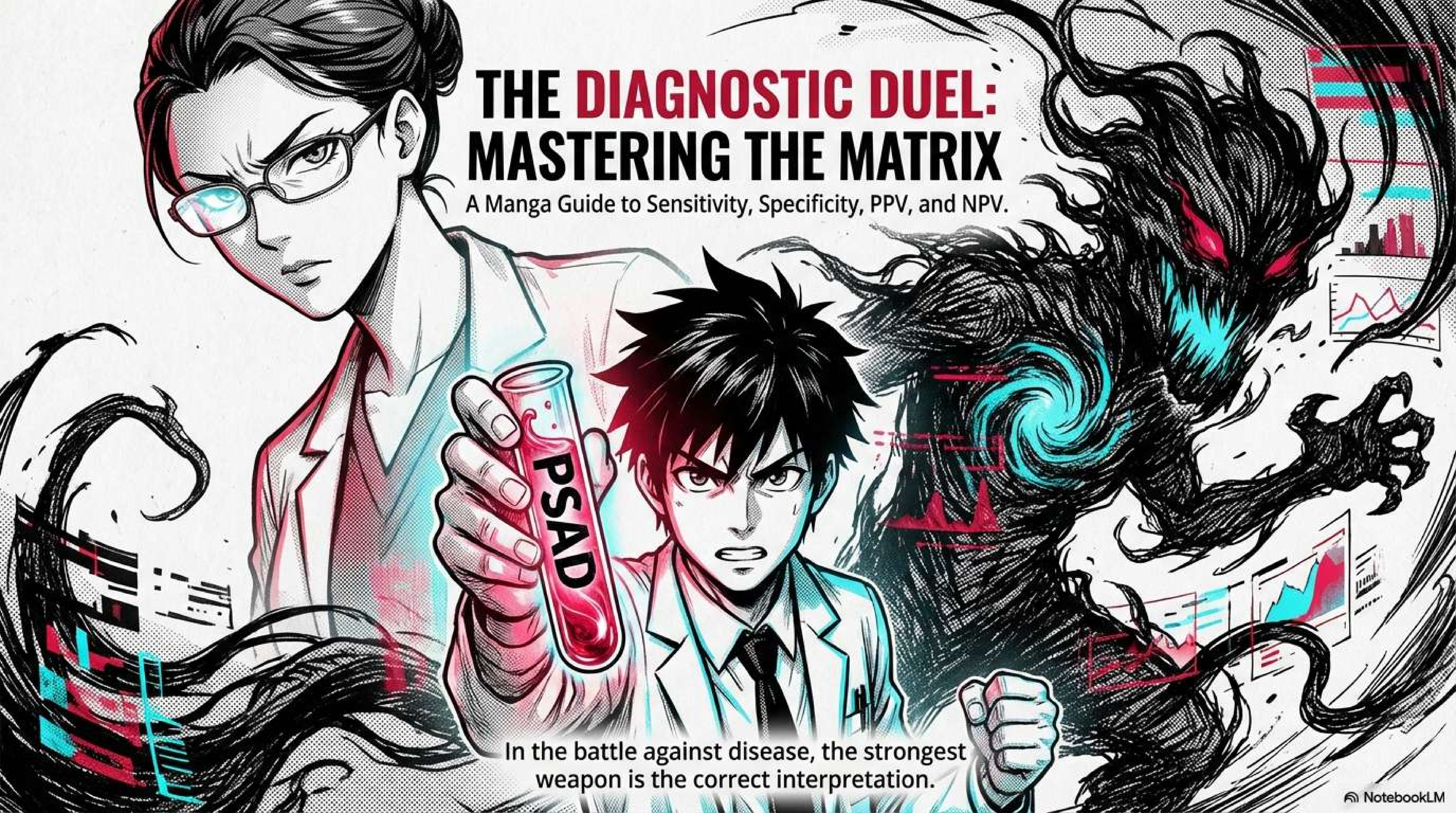
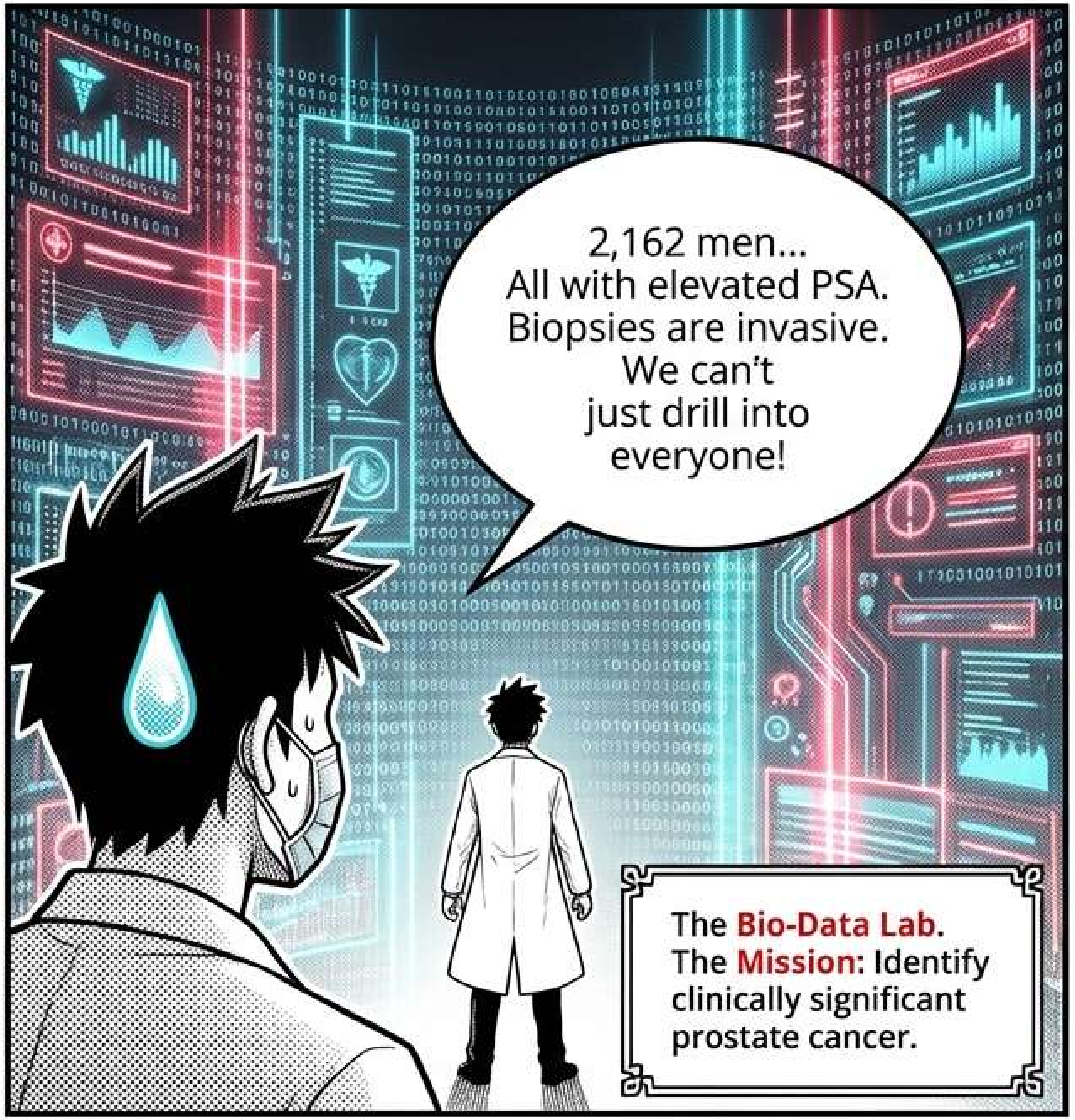
THE DIAGNOSTIC DUEL: MASTERING THE MATRIX

A Manga Guide to Sensitivity, Specificity, PPV, and NPV.

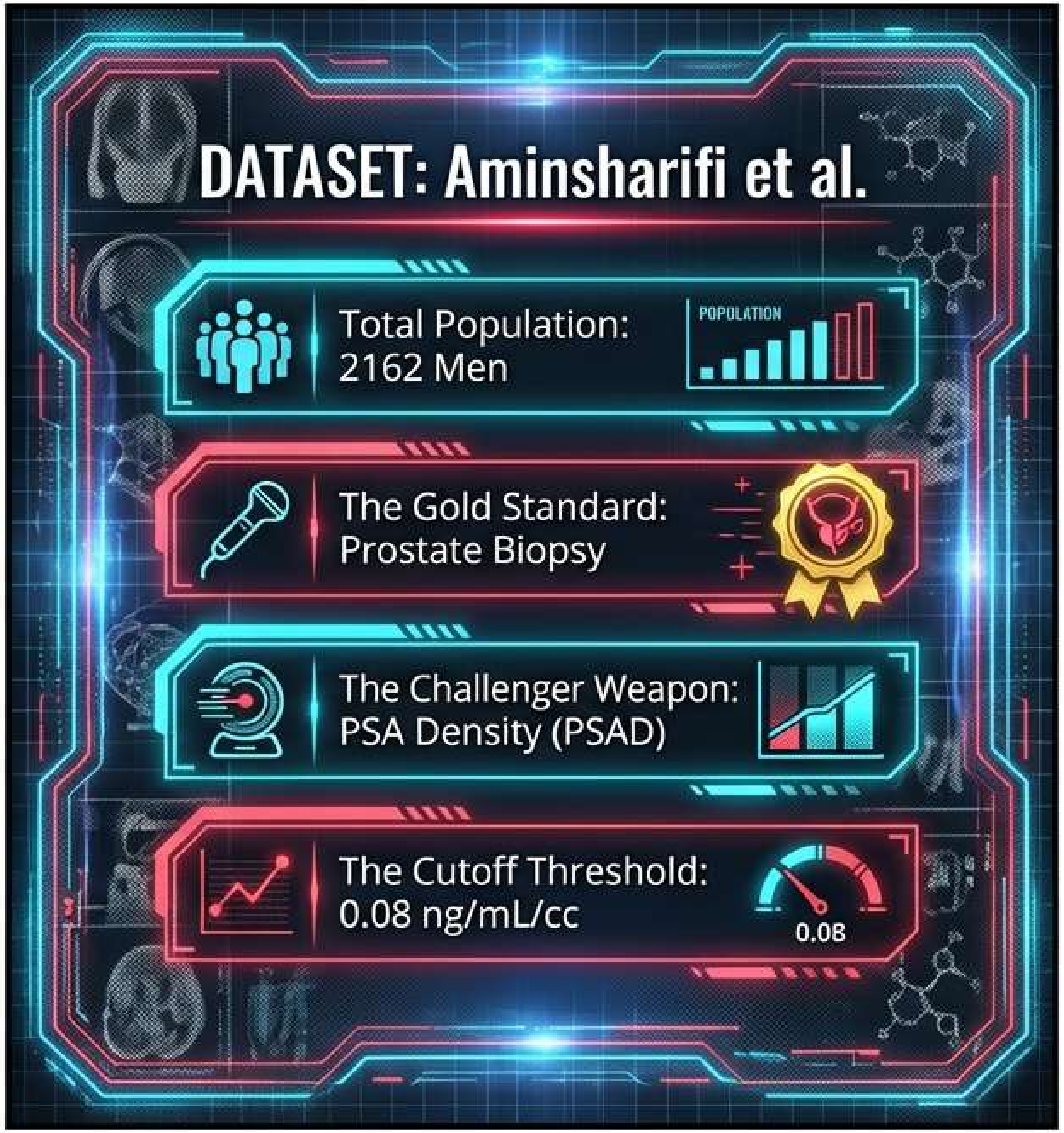


In the battle against disease, the strongest weapon is the correct interpretation.

NARRATIVE



DATA HUD



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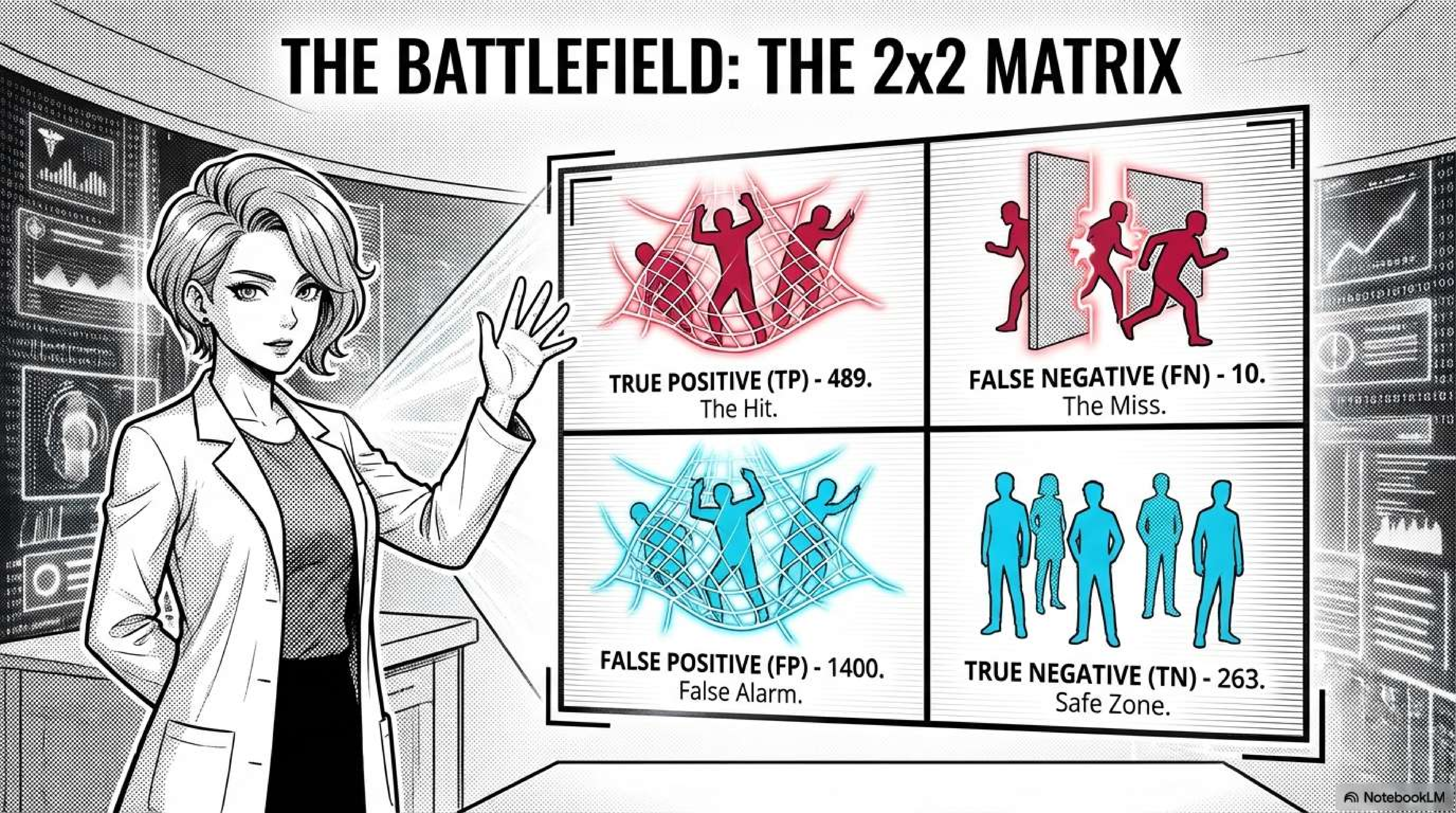
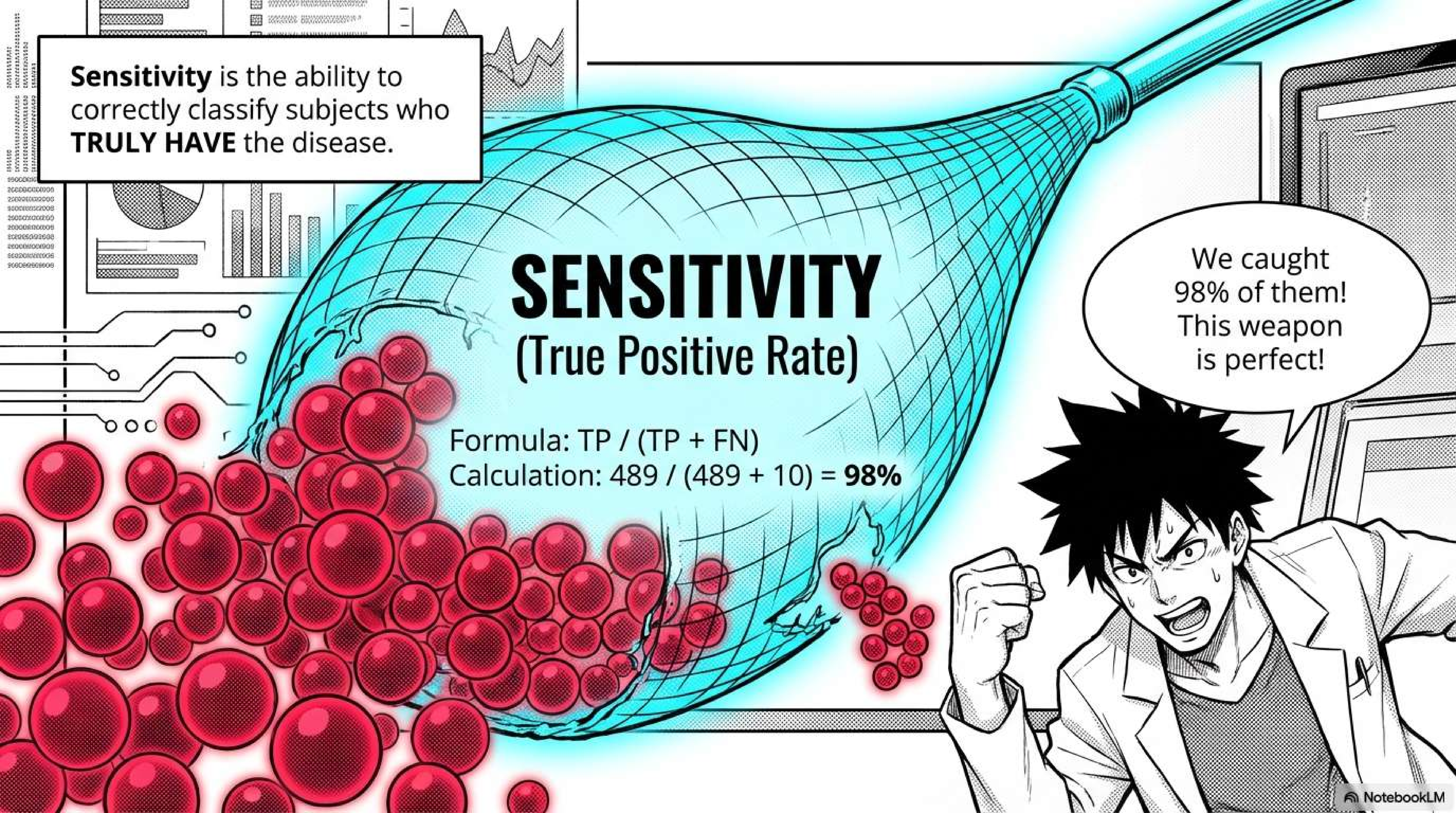

THE BATTLEFIELD: THE 2x2 MATRIX
TRUE POSITIVE (TP) - 489.
The Hit.
FALSE NEGATIVE (FN) - 10.
The Miss.
FALSE POSITIVE (FP) - 1400.
False Alarm.
TRUE NEGATIVE (TN) - 263.
Safe Zone.


Sensitivity is the ability to correctly classify subjects who **TRULY HAVE** the disease.
SENSITIVITY
(True Positive Rate)
Formula: TP / (TP + FN)
Calculation: 489 / (489 + 10) = **98%**
We caught 98% of them! This weapon is perfect!
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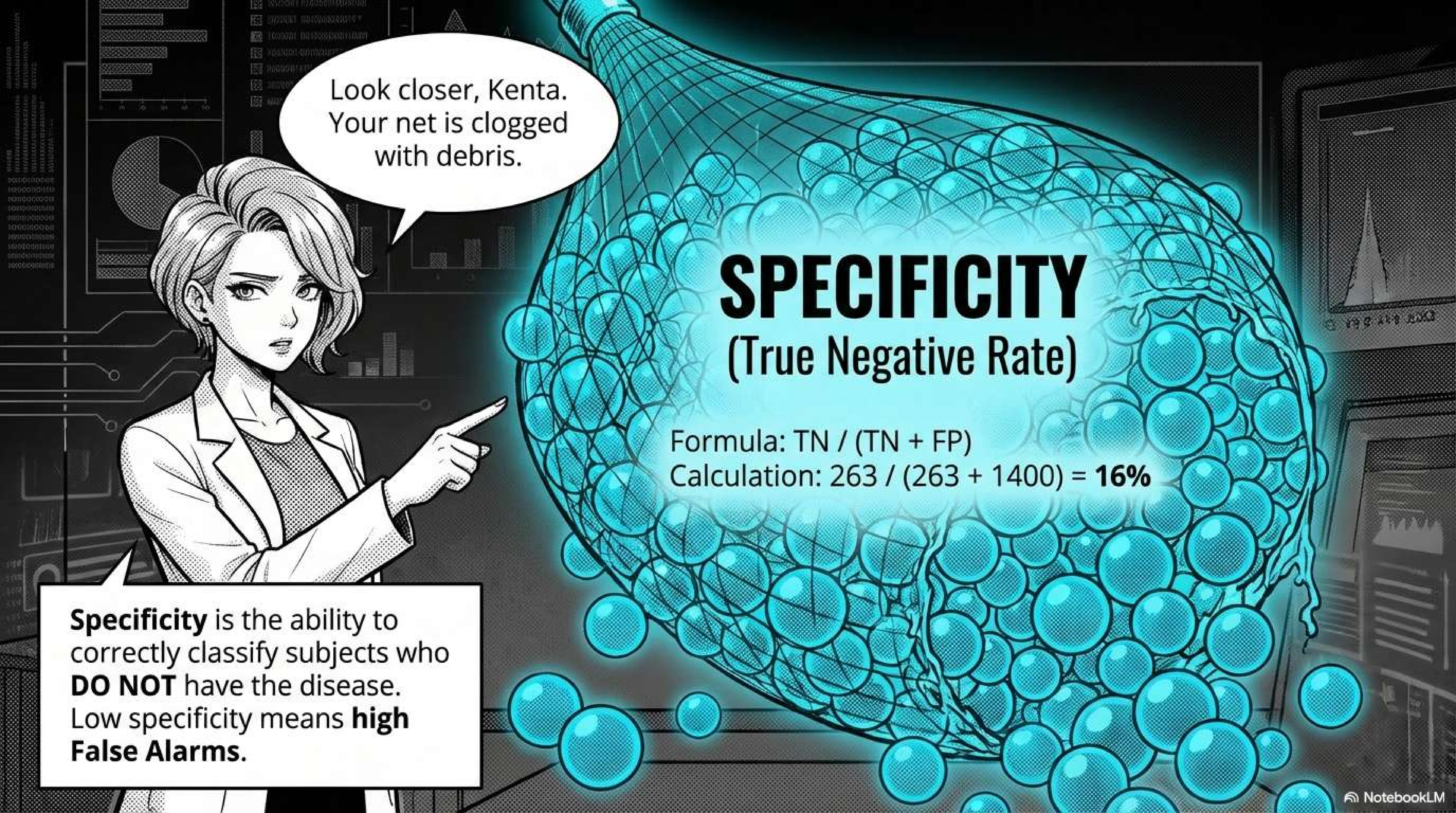
Look closer, Kenta. Your net is clogged with debris.
SPECIFICITY
(True Negative Rate)
Formula: TN / (TN + FP)
Calculation: 263 / (263 + 1400) = **16%**
Specificity is the ability to correctly classify subjects who **DO NOT** have the disease. Low specificity means **high False Alarms**.
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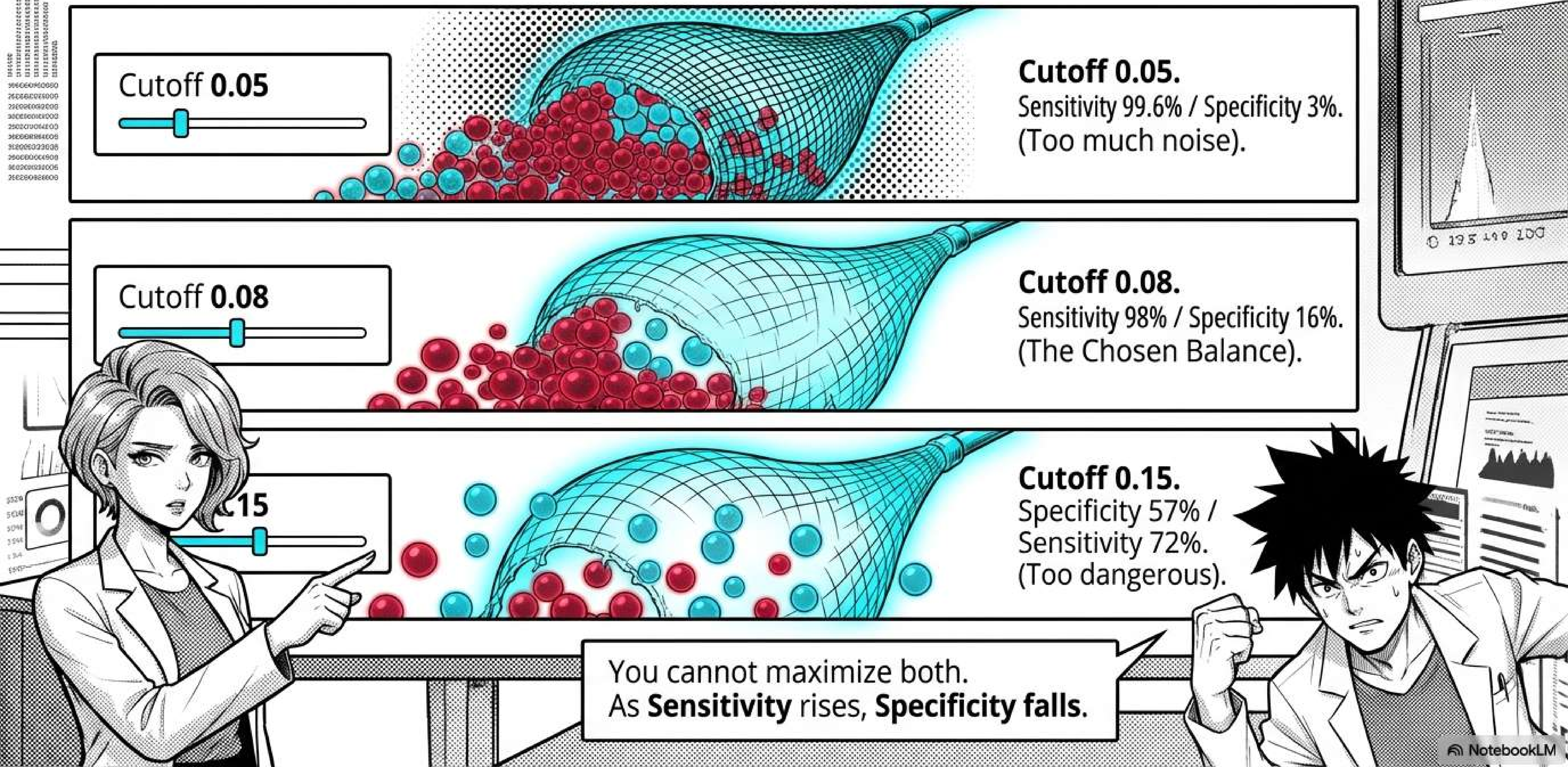
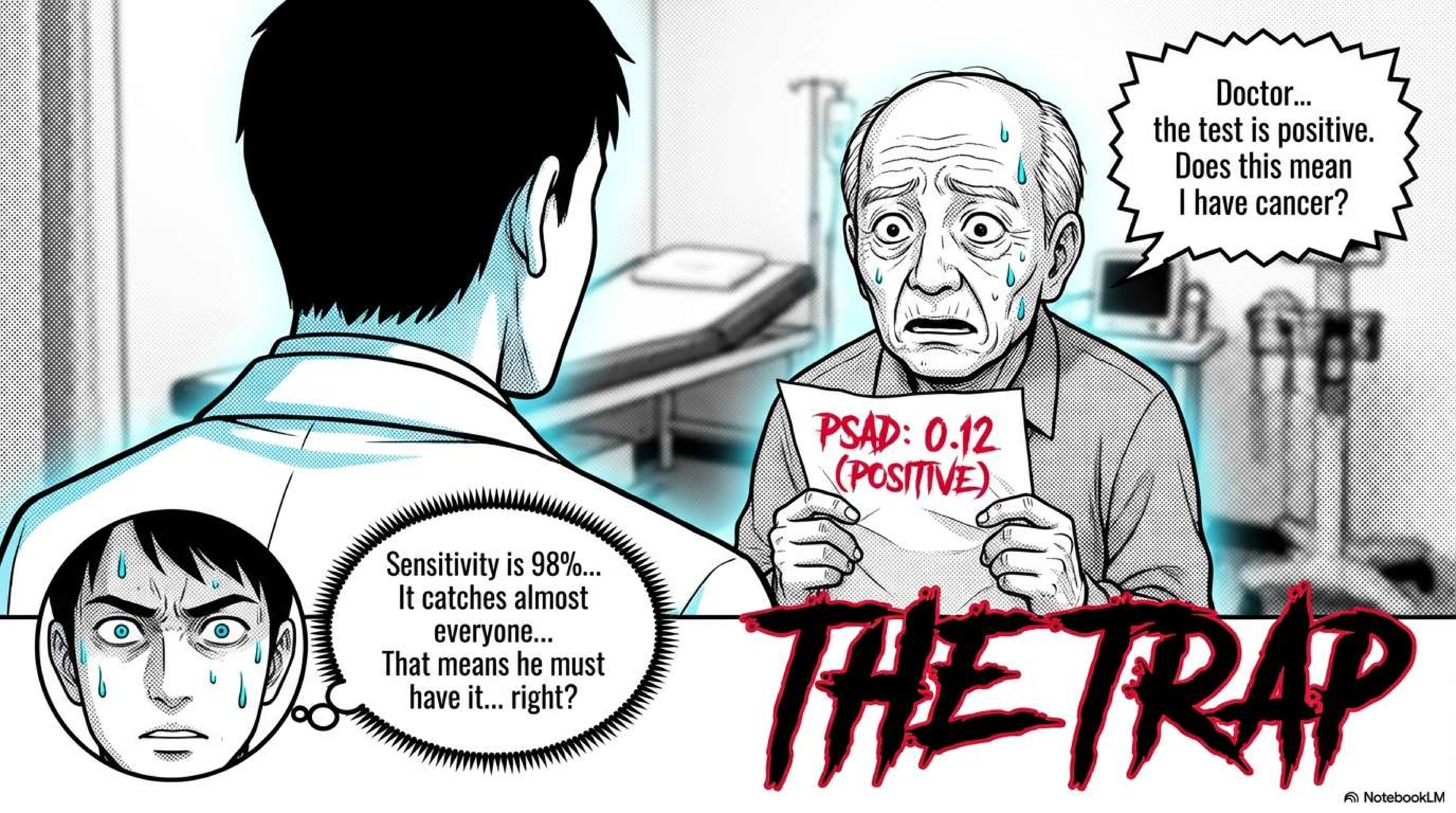
THE ETERNAL TRADE-OFF




Doctor... the test is positive. Does this mean I have cancer?
PSAD: 0.12 (POSITIVE)
Sensitivity is 98%... It catches almost everyone... That means he must have it... right?
THE TRAP
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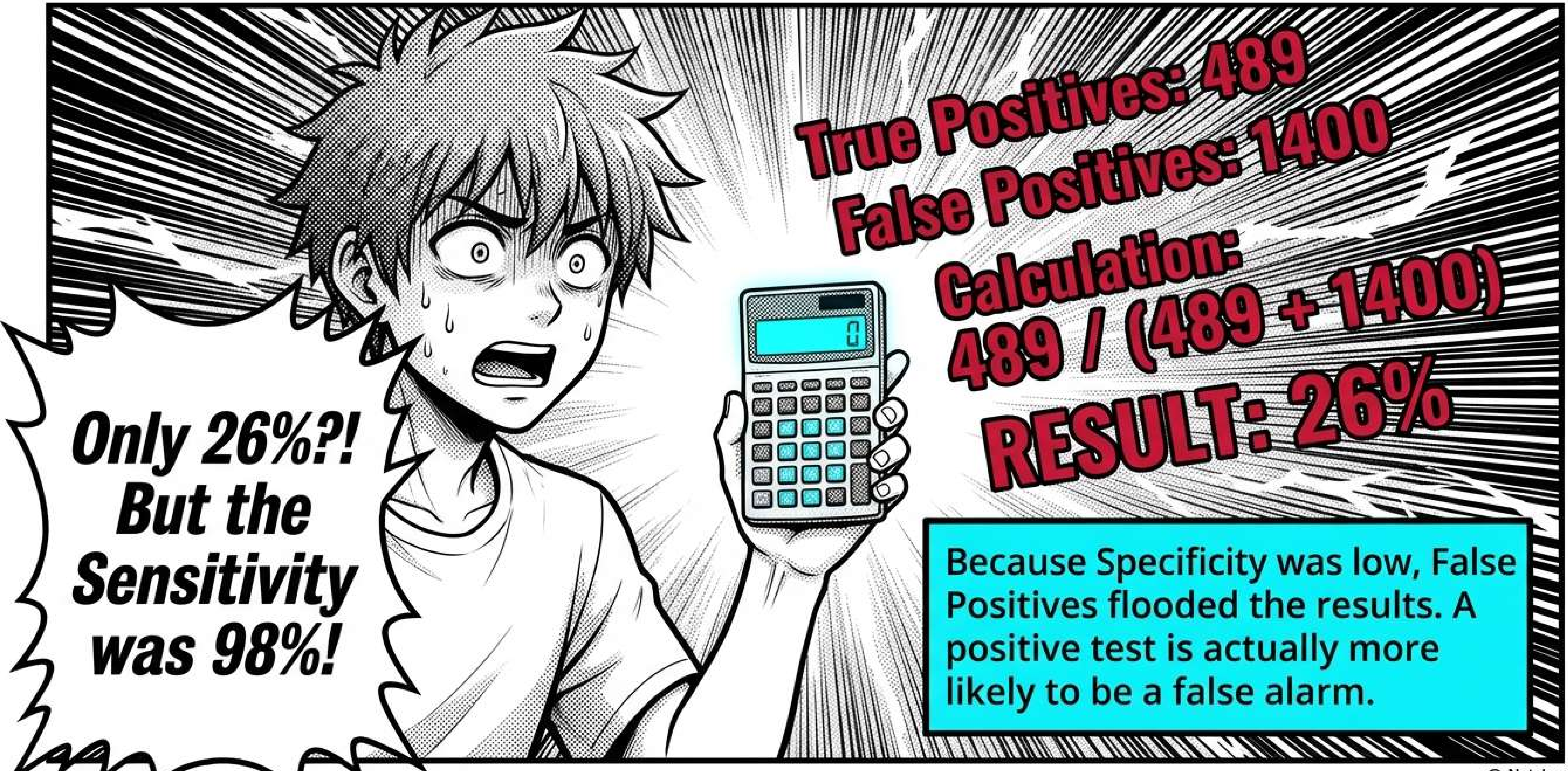
POSITIVE PREDICTIVE VALUE (PPV)

	DISEASE PRESENT	DISEASE ABSENT
TEST POSITIVE	TRUE POSITIVE (TP)	FALSE POSITIVE (FP)
TEST NEGATIVE	FALSE NEGATIVE (FN)	TRUE NEGATIVE (TN)

PPV: If the test is positive, what is the probability the patient actually has the disease?

Formula: TP / (TP + FP)

THE SHOCKING TRUTH



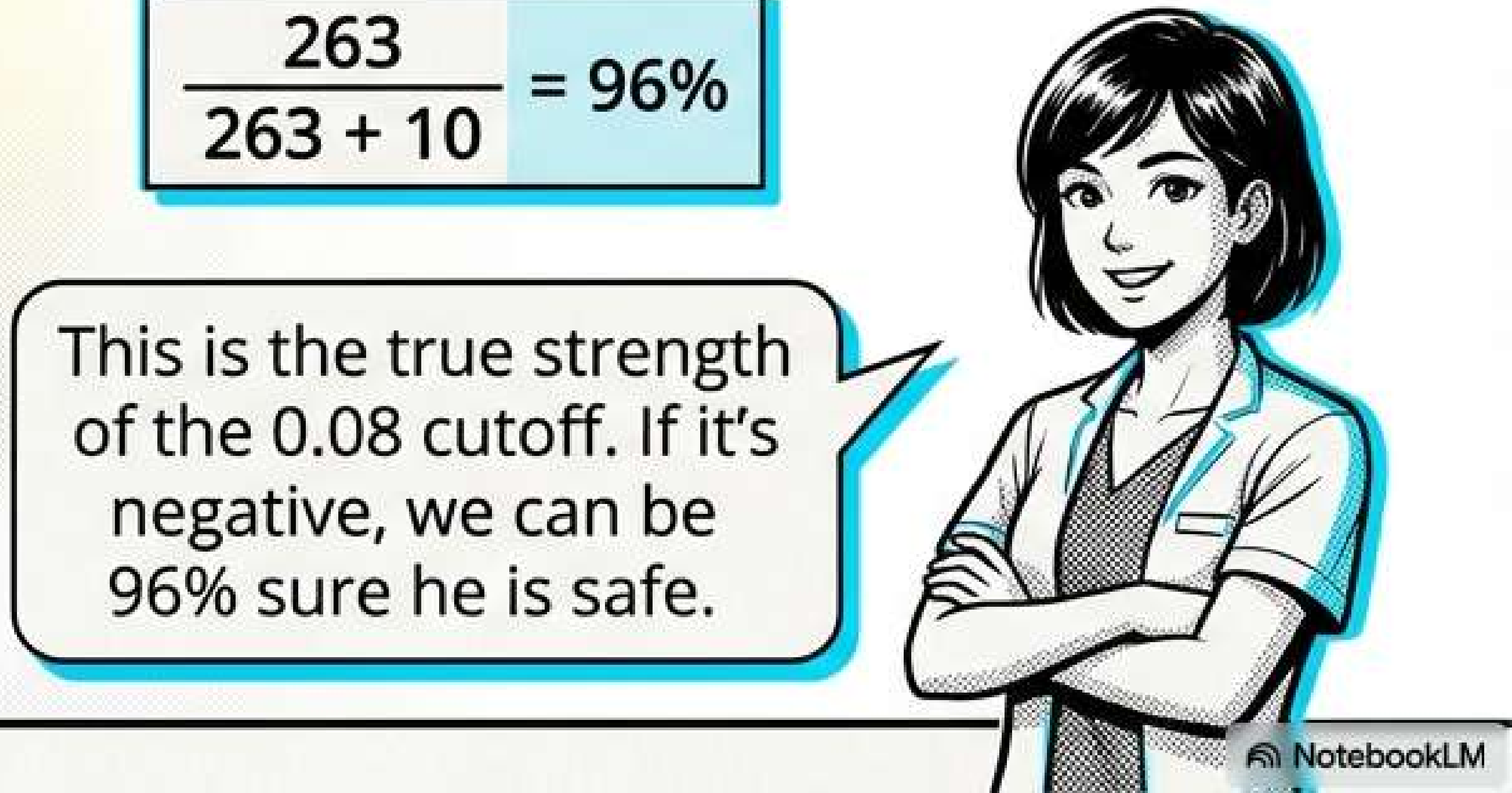
THE SAFETY NET: NPV

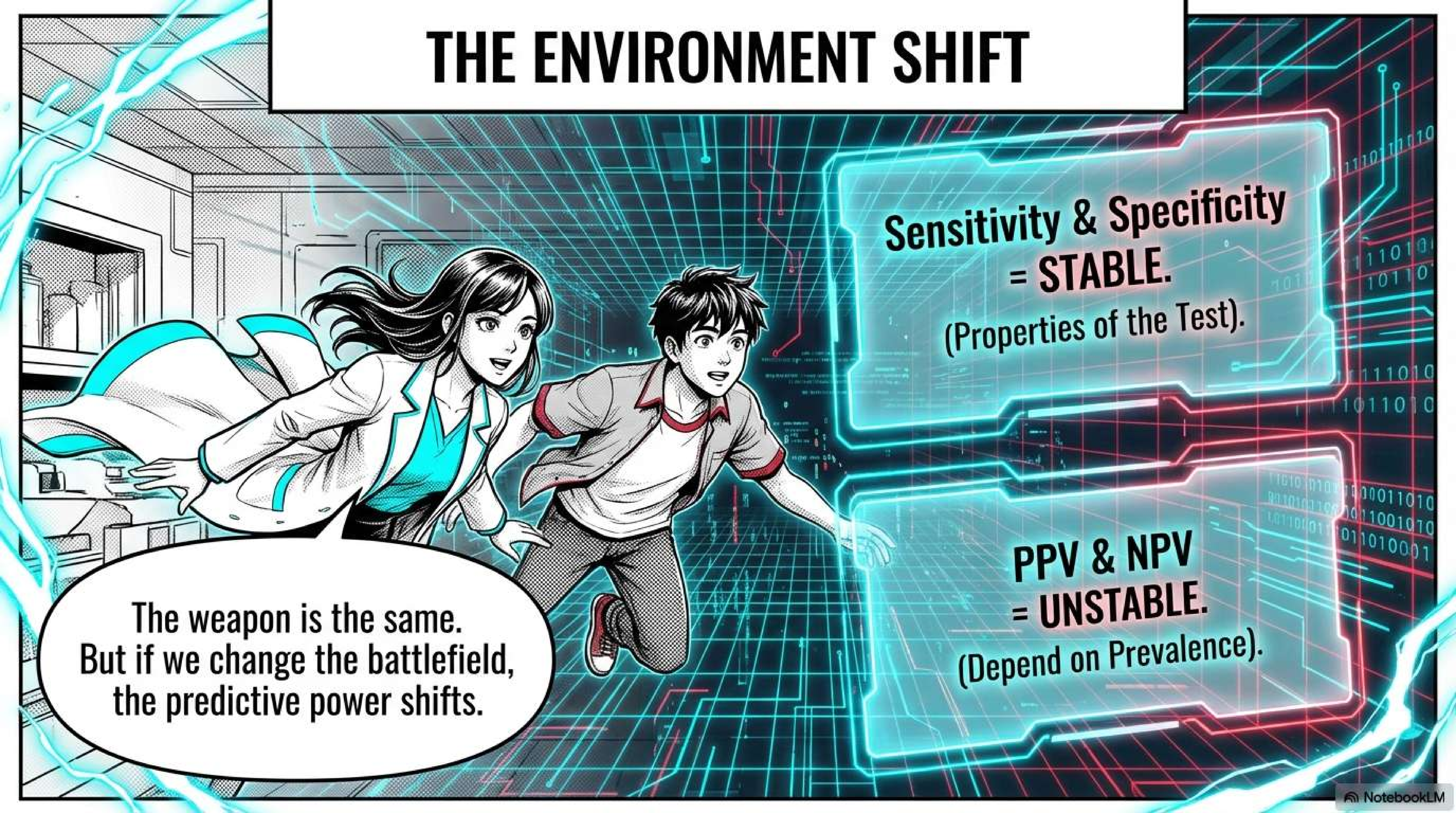
NEGATIVE PREDICTIVE VALUE (NPV)

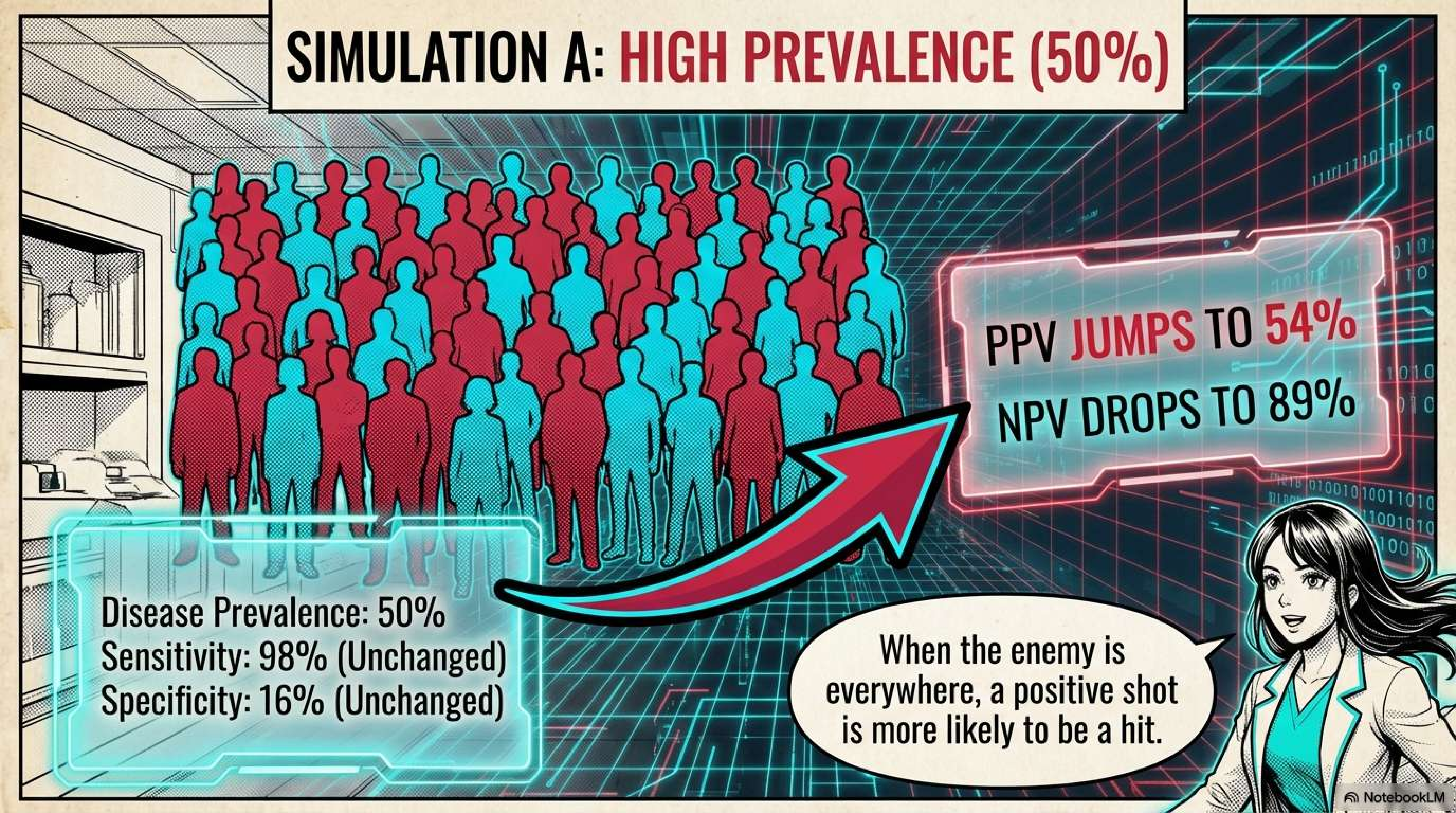
If the test is negative, what is the probability the patient is healthy?

$$\frac{TN}{TN + FN}$$

$$\frac{263}{263 + 10} = 96\%$$




THE ENVIRONMENT SHIFT
Sensitivity & Specificity = STABLE. (Properties of the Test).
PPV & NPV = UNSTABLE. (Depend on Prevalence).
The weapon is the same. But if we change the battlefield, the predictive power shifts.


SIMULATION A: HIGH PREVALENCE (50%)
PPV JUMPS TO 54%
NPV DROPS TO 89%
Disease Prevalence: 50%
Sensitivity: 98% (Unchanged)
Specificity: 16% (Unchanged)
When the enemy is everywhere, a positive shot is more likely to be a hit.
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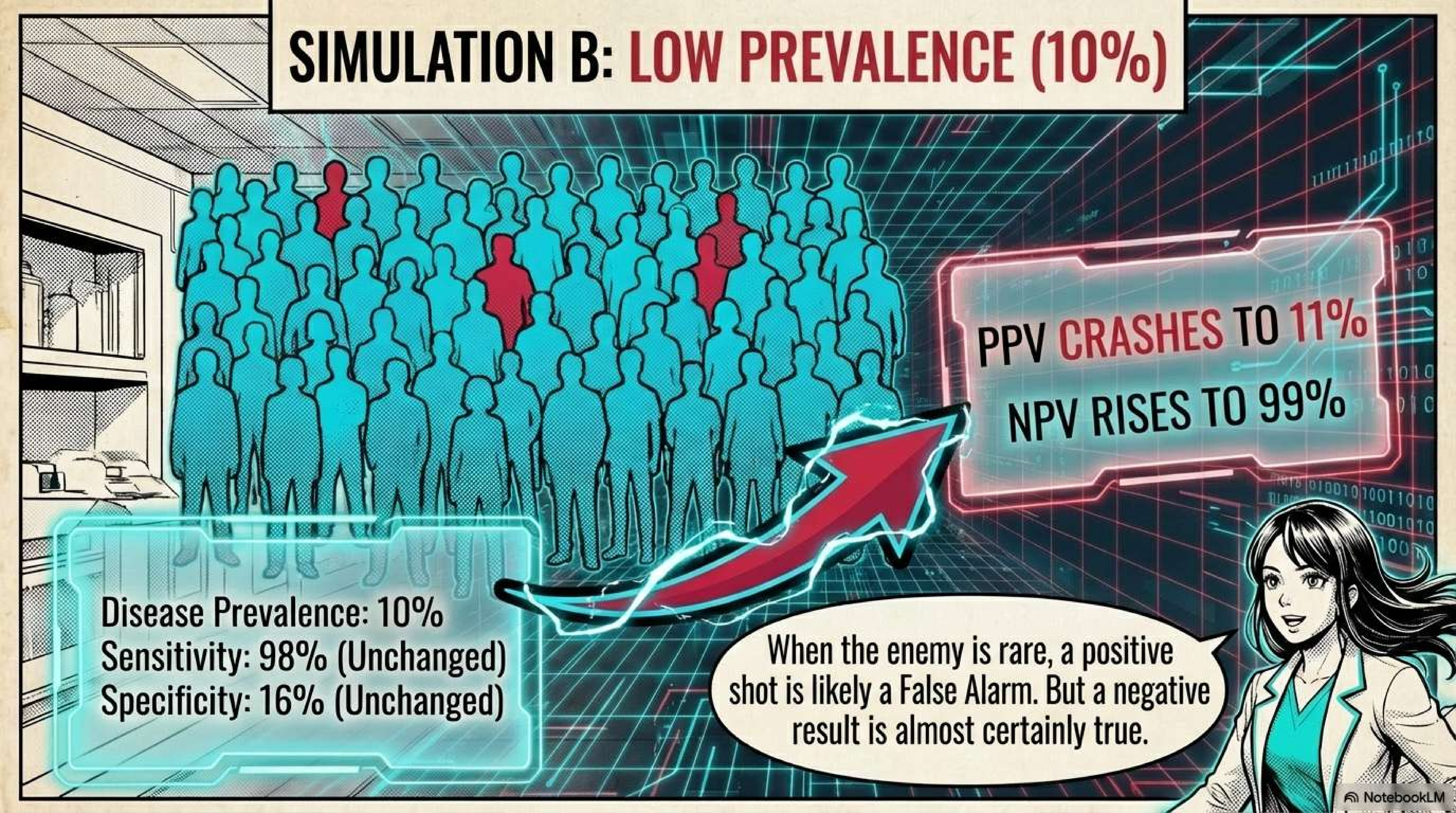
SIMULATION B: LOW PREVALENCE (10%)
PPV CRASHES TO 11%
NPV RISES TO 99%
Disease Prevalence: 10%
Sensitivity: 98% (Unchanged)
Specificity: 16% (Unchanged)
When the enemy is rare, a positive shot is likely a False Alarm. But a negative result is almost certainly true.
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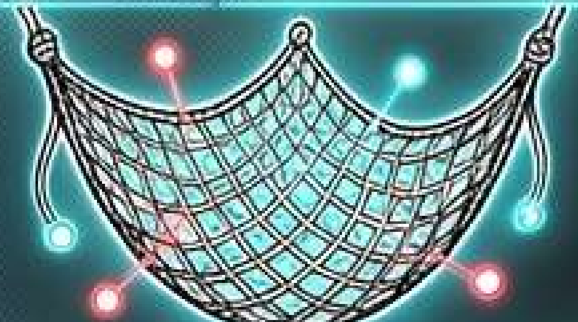
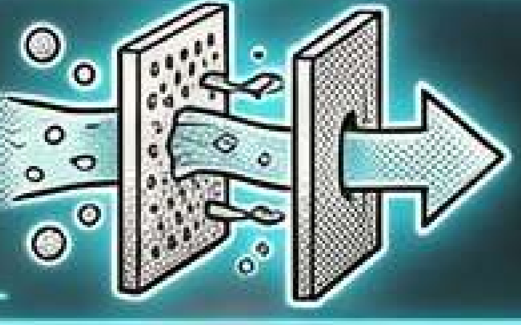
THE MASTER'S SCROLL



SENSITIVITY (TP Rate)
Finds the disease. Stable.



SPECIFICITY (TN Rate)
Identifies the healthy. Stable.

PPV (Trust the Yes)
Probability of disease if Positive.
Changes with Prevalence.

NPV (Trust the No)
Probability of health if Negative.
Changes with Prevalence.

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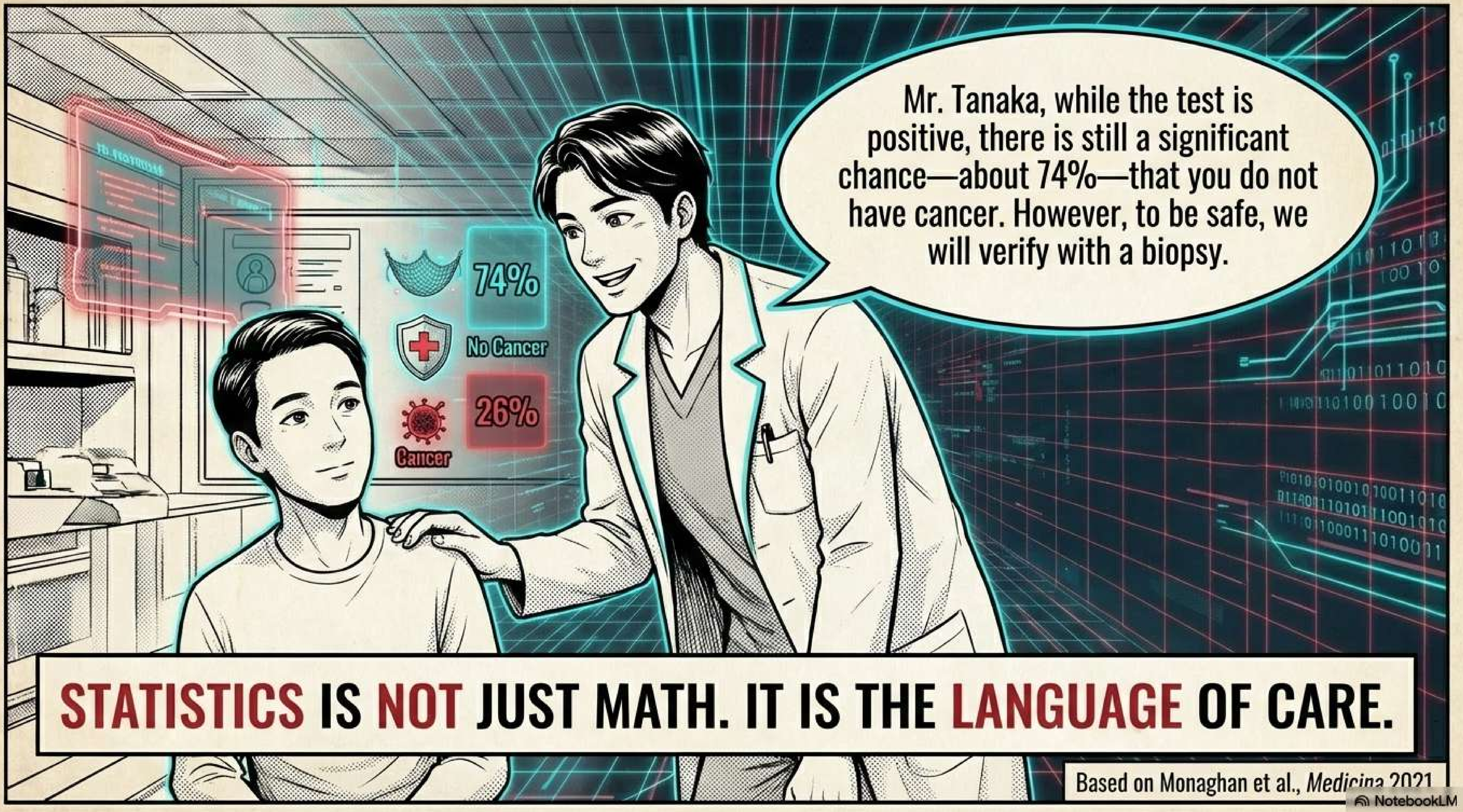
74%
No Cancer
26%
Cancer
Mr. Tanaka, while the test is positive, there is still a significant chance—about 74%—that you do not have cancer. However, to be safe, we will verify with a biopsy.
STATISTICS IS NOT JUST MATH. IT IS THE LANGUAGE OF CARE.
Based on Monaghan et al., *Medicina* 2021